Chapter 7

Exponential and Logarithmic Functions

7.1 Exponential Functions

We will begin by looking at exponential functions. Thereafter, we will look at logarithmic functions.

7.1.1 Laws of indices

Let $a, n, m \in \mathbb{R}$ with $a \neq 1$. We have the following;

1. $a^m \times a^n = a^{m+n}$. To multiply exponential expressions with the same base, copy the base and add the powers. For example, $5 \times 5^2 = 5^{1+2} = 5^3 = 125$.

2. $a^m \div a^n = a^{m-n}$. To divide exponential expressions with the same base, copy the base and subtract the indices. For example, $5(y^9 \div y^5) = 5y^{9-5} = 5y^4$.

3. $(a^m)^n = a^{mn}$.

4. $a^{-m} = \frac{1}{a^m}$. For example, $2^{-2} = \frac{1}{2^2} = \frac{1}{4}$.

5. $(a)^{\frac{m}{n}} = \sqrt[n]{a^m} = (\sqrt[n]{a})^m$

6. $a^0 = 1$

Definition 7.1.2. Let a be a positive real number that is not equal to 1. The exponential function with the base a is a function of the form

$$f(x) = a^x.$$

Example 7.1.3. Sketch the graphs of $f(x) = 3^x$ and $g(x) = \left(\frac{1}{3}\right)^x$ on the same Cartesian plane.

Solution

We begin by finding pairs (x, y) of values of x and y and then plot them.

x	-2	-1	0	1	2
$f(x) = 3^x$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9
$g(x) = \left(\frac{1}{3}\right)^x$	9	3	1	$\frac{1}{3}$	$\frac{1}{9}$

Finally we connect these points with a smooth curve, giving us the following graphs.

From the example above, we deduce the following.

1. If $y = a^x$, with $1 < a < \infty$, we see that as x becomes big, the corresponding value of y also becomes big. We also observe that if x is negative and x becomes smaller, y also becomes smaller. i.e y approaches 0. This is because if x is negative,

$$y = a^x = \frac{1}{a^{-x}},$$

which becomes very small as x approaches $-\infty$.

2. If $y = a^x$, with $0 < a < 1$,

we observe that as x becomes bigger and positive, then the corresponding values of y become smaller and approach zero. On the other hand, as x becomes bigger and negative the corresponding values of y become bigger.

The domain of the exponential function is $\mathbb{R}$, the set of real numbers, and its range is $(0, \infty)$.

The exponential function which is very important in most applications is the one whose base is e. The number e is an irrational number whose value is approximately 2.71828. This function is sometimes written exp x for exponential x. In fact whenever we speak of the exponential function without any reference to a base, it is understood that the base is e.

Example 7.1.4. Sketch the graph of each of the following on the same axes:

1. $y = 2^x$

2. $y = 3 + 2^x$

3. $y = -1 + 2^x$

From the example above, we deduce the following:

1. If $b > 0$, then the graph of $y = b + a^x$ is obtained from the graph of $y = a^x$ by shifting the graph of $y = a^x$, b units upwards.

2. If $b < 0$, then the graph of $y = b + a^x$ is obtained from the graph of $y = a^x$ by shifting the graph of $y = a^x$, b units downwards.

7.1.5 Exponential equations

Definition 7.1.6. Let a and b be positive real numbers with $a \neq 1$. A basic exponential equation is an equation of the form

$$a^x = b.$$

For example, $3^x = 81$ is an exponential equation. We solve a basic exponential equation by expressing both sides of the equation in terms of the same base or same powers.

Example 7.1.7. Solve each of the following:

1. $3^x = 81$

2. $x^3 = 125.$

3. $8^x = 0.125$

4. $\left(\frac{1}{8}\right)^{-x} = 2^{x+4}$.

Solutions

1.

$$\begin{aligned} 3^x &= 81 \\ 3^x &= 3^4 \\ x &= 4. \end{aligned}$$

2.

$$\begin{aligned} x^3 &= 125 \\ x^3 &= 5^3 \\ x &= 5. \end{aligned}$$

3.

$$\begin{aligned} 8^x &= 0.125 \\ 8^x &= \frac{125}{1000} \\ 8^x &= \frac{1}{8} \\ 8^x &= 8^{-1} \\ x &= -1. \end{aligned}$$

4.

$$
\begin{aligned}
\left(\frac{1}{8}\right)^{-x} &= 2^{x+4}\\
(8^{-1})^{-x} &= 2^{x+4}\\
8^{x} &= 2^{x+4}\\
(2^3)^{x} &= 2^{x+4}\\
2^{3x} &= 2^{x+4}\\
3x &= x+4\\
x &= 2.
\end{aligned}
$$

Some exponential equations are solved by first simplifying into basic exponential equations. To understand this, we have the following example.

Example 7.1.8. Solve each of the following:

1. $2^{2x} + 3(2^x) - 4 = 0$
2. $4^x - 2^{x+1} = 8$

Solutions

1.

$$
\begin{aligned}
2^{2x} + 3(2^x) - 4 &= 0\\
(2^x)^2 + 3(2^x) - 4 &= 0
\end{aligned}
$$

Let $p = 2^x$. We get

$$
\begin{aligned}
(2^x)^2 + 3(2^x) - 4 &= 0\\
p^2 + 3p - 4 &= 0,
\end{aligned}
$$

so that $p = 1$, and $p = -4$. If $p = 1$, then $2^x = 1$ is equivalent to $2^x = 2^0$, so that $x = 0$. If $p = -4$, then $2^x \neq -4$ since there is no such value of x which can give us -4. Therefore, $x = 0$ is the only solution.

2.

$$\begin{aligned} 4^x - 2^{x+1} &= 8 \\ (2^2)^x - 2^1 \times (2^x) - 8 &= 0 \end{aligned}$$

Let $p = 2^x$. We get

$$\begin{aligned} (2^2)^x - 2^1 \times (2^x) - 8 &= 0 \\ p^2 - 2p - 8 &= 0, \end{aligned}$$

so that $p = 4$, and $p = -2$. If $p = 4$, then $2^x = 4$ is equivalent to $2^x = 2^2$, so that $x = 2$. If $p = -2$, then $2^x \neq -2$ since there is no such value of x which can give us -2. Therefore, $x = 2$ is the only solution.

7.2 Logarithmic Functions

7.2.1 Basics about logarithms

Definition 7.2.2. Let a and x be positive real numbers with $a \neq 1$. The logarithm of x with base a is denoted by $\log_a x$ and is defined as the power to which a must be raised to obtain x. The function

$$y = f(x) = \log_a x,$$

is the logarithmic function with base a.

Definition 7.2.3. The function

$$y = f(x) = \log_e x = \ln x,$$

is the *natural* logarithmic function.

In the above definitions, when we write $\log x$ it is understood that the base is 10.

If $\log_a y = x$, then $a^x = y$, and if $a^y = x$, then $y = \log_a x$. Thus, the logarithm function, $y = \log_a x$, is the inverse of the exponential function $y = a^x$.

In the following examples, we will learn how to convert from exponential form to logarithmic form and vice versa.

Example 7.2.4. Write each of the following in Logarithmic form:

1. $3^2 = 9$
2. $\frac{1}{4} = 2^{-2}$
3. $8 = 2^{x+1}$

Solutions

1. If $3^2 = 9$, then $\log_3 9 = 2$.
2. Given $\frac{1}{4} = 2^{-2}$, we get $\log_2 \frac{1}{4} = -2$.
3. For $8 = 2^{x+1}$, we get $\log_2 8 = x + 1$.

Example 7.2.5. Write each of the following in exponential form:

1. $2 = \log_{10} 100$
2. $\log_3 x = 4$
3. $\log_x 64 = 3$

Solutions

1. If $2 = \log_{10} 100$ then $10^2 = 100$.

2. If $\log_3 x = 4$, then $3^4 = x$.

3. If $\log_x 64 = 3$, then $64 = x^3$.

7.2.6 Rules of logarithms

The rules of logarithms are similar to the rules for indices. Let $P = a^m$ and $Q = a^n$, then $m = \log_a P$ and $n = \log_a Q$. It follows that

1. $PQ = a^m \times a^n = a^{m+n}$, so that $\log_a PQ = m + n = \log_a P + \log_a Q$.

2. $\frac{P}{Q} = \frac{a^m}{a^n} = a^{m-n}$, and so $\log_a \frac{P}{Q} = m - n = \log_a P - \log_a Q$.

3. $P^n = (a^m)^n = a^{mn}$. Thus, $\log_a P^n = nm = n \log_a P$.

Note

1. $\log_a 1 = 0$, this is because $a^0 = 1$.

2. $\log_a a = 1$ this is because $a^1 = a$.

3. $\log_a x$ is not defined if $x < 0$. The logarithm of a negative number does not exist.

4. $0 < x < 1$, then $\log_a x < 0$. The logarithm of a number less than 1 is negative.

5. $\log_a 0$ is an defined.

6. As x increases, $\log_a x$ increases.

7. Given $\log_b N$, we may wish to change to another base say base a.

   Proof. Let $\log_b N = y$, then $b^y = N$, so that $\log_a b^y = \log_a N$. It follows that $y \log_a b = \log_a N$, and so $y = \frac{\log_a N}{\log_a b}$. Therefore, from $\log_b N = y$, and $y = \frac{\log_a N}{\log_a b}$, we get $\log_b N = \frac{\log_a N}{\log_a b}$. □

   For example, $\log_2 8 = \frac{\log_8 8}{\log_8 2}$.

Example 7.2.7. Given that $\log_5 2 = 0.431$ and $\log_5 3 = 0.683$. Find

1. $\log_5 6$
2. $\log_5 1.5$
3. $\log_5 8$
4. $\log_5 12$
5. $\log_5 \frac{1}{18}$.

Solutions The general ideal here is to use the rules of logarithms to express the given 'logarithm' in terms of 2 and 3 whose values we know.

1. $\log_5 6 = \log_5(2 \times 3) = \log_5 2 + \log_5 3 = 0.431 + 0.683 = 1.114$
2. $\log_5 1.5 = \log_5(\frac{3}{2}) = \log_5 3 - \log_5 2 = 0.683 - 0.431 = 0.252$
3. $\log_5 8 = \log_5 2^3 = 3\log_5 2 = 3(0.431) = 1.293$
4. HW
5. HW

Example 7.2.8. Simplify each of the following:

1. $\log_{10} 100$
2. $\log_2 8$
3. $\log_3 \frac{1}{9}$

Solutions

1. $\log_{10} 100 = \log_{10} 10^2 = 2\log_{10} 10 = 2 \times 1 = 2$
2. $\log_2 8 = \log_2 2^3 = 3\log_2 2 = 3 \times 1 = 3.$
3. $\log_3 \frac{1}{9} = \log_3 1 - \log_3 9 = 0 - 2 = -2$

7.2.9 Basic Logarithmic equation

A basic logarithmic equation is an equation of the form

$$\log_a b = x \text{ or } \log_x b = a \text{ or } \log_a x = b,$$

where $a, b, x \in \mathbb{R}$ obeying the rules defined in the previous section.

We solve a basic log. equation by first converting it to exponential form, and then solve the way we did with exponential equations.

Example 7.2.10. Solve each of the following:

1. $x = \log_2 64$
2. $\log_x 8 = 3$
3. $\log_{216} x = \frac{1}{3}$

Solutions

1.

$$\begin{aligned} x &= \log_2 64 \\ 2^x &= 64 \\ 2^x &= 2^6 \\ x &= 6. \end{aligned}$$

2.

$$\begin{aligned} 3 &= \log_x 8 \\ x^3 &= 8 \\ x^3 &= 2^3 \\ x &= 2. \end{aligned}$$

3. HW

Example 7.2.11. Solve each of the following:

1. $2\log_5 x = \log_5(2x+3)$

2. $\log_3(x^2+2) = 1 + \log_3(x+2)$

Solutions

1.

$$\begin{aligned}
2\log_5 x &= \log_5(2x+3)\\
\log_5 x^2 &= \log_5(2x+3)\\
\log_5 x^2 - \log_5(2x+3) &= 0\\
\log_5\left(\frac{x^2}{2x+3}\right) &= 0\\
\frac{x^2}{2x+3} &= 5^0\\
\frac{x^2}{2x+3} &= 1\\
x^2 - 2x - 3 &= 0\\
x = 3, x = -1.
\end{aligned}$$

Since $2\log_5(-1)$ is not defined, $x = 3$.

2.

$$\begin{aligned}
\log_3(x^2+2) &= 1 + \log_3(x+2)\\
\log_3(x^2+2) &= \log_3 3 + \log_3(x+2)\\
\log_3(x^2+2) &= \log_3(3(x+2))\\
x^2 + 2 &= 3x + 6\\
x^2 - 3x - 4 &= 0\\
x = 4, x = -1.
\end{aligned}$$

Both values satisfy the given equation.

Certain logarithmic equations are solved by first converting one base to the other and then solve as before. We demonstrate this notion in the next example.

Example 7.2.12. Solve each of the following:

1. $\log_2 x + \log_4 x = 6$.

2. $\log_2 x + \log_4 x + \log_{16} x = 7$.

3. $\log_x 2 - \log_4 x + \frac{7}{6} = 0$.

Solutions

1.

$$\begin{aligned}
\log_2 x + \log_4 x &= 6 \\
\log_2 x + \frac{\log_2 x}{\log_2 4} &= 6 \\
\log_2 x + \frac{\log_2 x}{2\log_2 2} &= 6 \\
2\log_2 x + \log_2 x &= 12 \\
\log_2(x(x^2)) &= 12 \\
x^3 &= 2^{12} \\
x^3 &= (2^4)^3 \\
x &= 2^4 = 16.
\end{aligned}$$

2.

$$\begin{aligned}
\log_2 x + \log_4 x + \log_{16} x &= 7 \\
\log_2 x + \frac{\log_2 x}{\log_2 4} + \frac{\log_2 x}{\log_2 16} &= 7 \\
\log_2 x + \frac{\log_2 x}{2\log_2 2} + \frac{\log_2 x}{4\log_2 2} &= 7 \\
4\log_2 x + 2\log_2 x + \log_2 x &= 28 \\
\log_2 x^4 + \log_2 x^2 + \log_2 x &= 28 \\
\log_2 x^7 &= 28 \\
2^{28} &= x^7 \\
x &= 16.
\end{aligned}$$

3. HW

Equations in exponential form which can not be expressed so that the bases or exponents are the same are easily solved by introducing the natural logarithm, as shown below.

Example 7.2.13. Solve each of the following.

1. $2^{x+1} = 3$

2. $\log_3 4 = x$

Solutions

1.

$$\begin{aligned} 2^{x+1} &= 3 \\ \ln 2^{x+1} &= \ln 3 \\ (x+1)\ln 2 &= \ln 3 \\ x\ln 2 + \ln 2 &= \ln 3 \\ x &= \frac{\ln 3 - \ln 2}{\ln 2} \\ x &= \frac{\ln 1.3}{\ln 2}. \end{aligned}$$

2.

$$\begin{aligned} \log_3 4 &= x \\ 3^x &= 4 \\ \ln 3^x &= \ln 4 \\ x\ln 3 &= \ln 4 \\ x &= \frac{\ln 4}{\ln 3}. \end{aligned}$$

7.2.14 Graphs of the logarithmic functions

As the logarithmic function, $y = \log_a x$, is the inverse of the exponential function $x = a^y$, we can obtain its graph by reflecting $y = a^x$ in the line $y = x$.

In the next example, we will sketch the graphs of logarithmic functions by first converting the function to exponential form, and then sketch as before.

Example 7.2.15. Sketch the graph of each of the following:

1. $y = \log_2 x$
2. $y = \log_{\frac{1}{2}} x$

7.2.16 Applications of exponential functions

The following example demonstrates the applications of the concept of exponential functions in real life situations. The solutions to this example has , however, been left as home work (HW).

Example 7.2.17.

1. Because of a slump in the economy, a company finds that its annual revenues have dropped from K742000 in 2014 to K632000 in 2016. If the revenue is following an exponential pattern of decline, what is the expected revenue for 2017?
2. The amount $A(t)$, in grams, of a certain radioactive substance remaining after t years decays by the formula $A(t) = A_0 e^{-0.0045t}$, where A_0 is the initial amount.
   (a) If 5 grams are left after 800 years, how many grams were present initially?
   (b) What is the half-life of the substance?

NOTE: THESE LECTURE NOTES ON EXPONENTIAL AND LOGARITHMIC FUNCTIONS ARE MEANT FOR MAT1110 (2020/2021) ACADEMIC YEAR ONLY.